Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 12)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 1 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

0 1 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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0 2 . 1 Give **three** advantages of incinerating domestic waste.

[3 marks]

1 .....

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2 .....

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3 .....

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0 2 . 2 Explain how each of the following features of a modern incinerator reduces pollution.

[4 marks]

High combustion temperature with rapid cooling of gases .....

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Water spray scrubber .....

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Removal of wet / compostable waste .....

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Removal of recyclable waste .....

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0 2 . 3 Explain why the ash from waste incinerators needs careful disposal.

[2 marks]

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0 2 . 4 State **one** reason why incineration is expensive.

[1 mark]

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0 3 . 1 Give **three** sources of mercury pollution.

[3 marks]

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0 3 . 2 Describe **three** methods used to control mercury pollution.

[3 marks]

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2 .....

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0 3 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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0 3 . 4 State what is meant by synergism and give an example involving a heavy metal.

[2 marks]

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0 4 . 1 Describe polymetallic (manganese) nodules and explain why they are not yet exploited on a large scale.

[3 marks]

0 4 . 2 Explain the environmental impacts of mining polymetallic nodules.

[3 marks]

0 4 . 3 Explain why rare earth metals are a supply concern and describe one new extraction technique.

[4 marks]

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0 5 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 5 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 5 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 6 Table 1 shows some properties of four groups of insecticide.

Table 1

Insecticide group	Toxicity to insects	Toxicity to mammals	Persistence	Liposolubility
Organochlorines (eg DDT)	High	Low	High	High
Organophosphates (eg malathion)	High	High	Low	Low
Pyrethroids	High	Low	Low	Low
Neonicotinoids	High	Low	Moderate	Low (water soluble)

0 6 . 1 Use Table 1 to explain why organochlorine insecticides caused the decline of birds of prey such as the peregrine falcon in the 1950s and 1960s.

[4 marks]

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0 6 . 2 Organophosphates replaced organochlorines for many uses. Use Table 1 to give **one** advantage and **one** disadvantage of this change.

[2 marks]

Advantage .....

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Disadvantage .....

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0 6 . 3 Explain why pyrethroids should not be used near rivers or fish farms.

[1 mark]

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0 6 . 4 Describe how the properties of neonicotinoids allow them to be used as a seed coating, and explain why this method of application reduces harm to non-target species.

[3 marks]

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0 7 A 1960s office block is being refurbished. Table 1 shows the measured heat loss through its fabric before the work.

Table 1

Part of the building	Heat loss / kW
Walls (uninsulated)	42
Roof	26
Single-glazed windows	31
Floor	9
Draughts and ventilation	22

0 7 . 1 Calculate the percentage of the total heat loss that passes through the windows.

[2 marks]

Answer ..... %

0 7 . 2 Explain how each of the following reduces heat loss.

[3 marks]

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Cavity wall insulation .....

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Low-emissivity double glazing .....

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A heat exchanger on the ventilation system .....

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0 7 . 3 Explain why there is an optimum thickness of loft insulation rather than simply using as much as possible.

[2 marks]

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0 7 . 4 The refurbishment also includes a combined heat and power (CHP) unit. Explain why a CHP unit has a lower **electrical** efficiency than a large power station but a much higher **overall** efficiency.

[3 marks]

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0 7 . 5 Explain **five** other ways the refurbished building could be designed or managed to use less energy.

[5 marks]

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0 8 . 1 Explain how each of the following human activities alters processes in the hydrological cycle.

[6 marks]

Deforestation

Urban development

Agriculture

0 8 . 2 Describe **two** ways in which global climate change alters the hydrological cycle.

[2 marks]

1 .....

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2 .....

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0 8 . 3 Explain why the impact of industrial and domestic water use on the hydrological cycle is usually local.

[2 marks]

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0 9 . 1 Complete Table 1 to show the source and disposal method for each category of radioactive waste from nuclear electricity generation.

[6 marks]

Table 1

Waste category	Example / source	Storage or disposal method	
High-level waste			
Intermediate-level waste			
Low-level solid waste			

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0 9 . 2 Explain why high-level waste must be cooled and why its storage must be monitored for very long periods.

[2 marks]

0 9 . 3 Describe how low-level liquid and gaseous wastes are treated before discharge.

[2 marks]

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1 0 . 1 Using named examples, explain how the sources of noise pollution can be controlled by changes to design, operation and location.

[9 marks]

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Extra space

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1 0 . 2 State what is meant by the **Leq 57 dB** contour used around UK airports.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

OR

11 . 2 Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS